

Ákos Makra, PhD

Test Automation Engineer

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SUMMARY

Test automation engineer with 4 years of testing experience and a PhD in Information Science. At Magyar Telekom, writes and maintains automated UI, E2E and system regression tests in Python, based on business requirements. Led the effort to fix unstable regression tests on two projects, raising their pass rate from under 65% to over 95%. Previously wrote unit and integration tests for safety-critical dialysis machine software at B. Braun.

SKILLS

Test Automation: Playwright, Selenium, Behave/Gherkin (BDD), CppUTest; UI, E2E, system, regression, unit and integration testing

Programming: Python, C, C++, SQL, MATLAB

CI/CD & Version Control: GitLab CI/CD, Git

API Testing: REST API tests in Python, Postman

AI-Assisted Development: Claude Code

Tools: Jira, Zephyr, Confluence, Kibana, OpenSearch, IBM DOORS, Ruff

Languages: English (fluent), German (intermediate), Hungarian (native)

WORK EXPERIENCE

Test Automation Engineer

Jul 2024 – Present

Magyar Telekom Nyrt.

- Maintain and extend the team's 450+ test regression suite (Python, Behave, Selenium/Playwright) for the customer service agents' application and its order flows.
- Led regular team sessions for 9 months to fix unstable regression tests on two projects. Main causes: faulty test logic (wrong assertions, inconsistent waits, hard-coded data) and test data that was shared or poorly defined for the given process. Fixed the logic and refined the test data. Pass rate went from under 65% to over 95%.
- Together with automating 30% of the manual cases, this cut regression on the largest project from 6 testers over 3–4 days to 3 testers in one day.
- Refactor test code into clear layers (page objects, step definitions, waits and assertions). Removed several thousand lines of unused code, including dependencies in other repositories.
- Set up scheduled GitLab CI/CD pipelines. Investigate failed runs in Kibana/OpenSearch and report results in Zephyr.
- Proved with logs that the test environment failed under parallel load and followed up until it was fixed, nearly halving automated test runtime.
- Use Claude Code for refactoring and new test cases, with review agents and hooks for code checks.
- Mentor manual testers and new colleagues in automation and log-based debugging. Interview test automation candidates.

Software Tester (embedded C/C++)

Nov 2021 – Jun 2023

B. Braun (via Effective Group Talent Solutions Kft.)

- Wrote automated unit and integration tests in C and C++ (CppUTest) for safety-critical software on acute dialysis machines.
- Worked through a years-old backlog of integration and unit tests, raising code coverage.
- Checked code against requirements in IBM DOORS and reported mismatches.

Researcher

May 2019 – Sep 2019

MedRes Kft.

- Ran GPU-based ultrasound simulations in k-Wave (MATLAB) and helped with medical device development and testing.

EDUCATION

PhD, Information Science · Roska Tamás Doctoral School of Sciences and Technology

2015 – 2020

Thesis: Ultrasound Image Modelling and Resolution Enhancement

MSc, Info-Bionics Engineering · Pázmány Péter Catholic University

2014 – 2015

BSc, Molecular Bionics Engineering · Pázmány Péter Catholic University

2011 – 2014

Research internships (Erasmus): Philips, Eindhoven, NL (2014) · University of Florence, MSDLAB, IT (2015) · Fraunhofer IBMT, DE (2016)

Scholarships & awards: New National Excellence (2018, 2019), Young Talent (2017), PhD Excellence (2016), Central Fund (2015) · 1st place, Student Scientific Conference (2014, 2015)

PERSONAL ACHIEVEMENTS

- First Hungarian to thru-hike the 4,273 km Pacific Crest Trail southbound (130 days, 2023) and the 800 km Pyrenean Haute Route (40 days, 2021).